

LeadX Pro AI

Lead Intelligence & Business Data Platform

Project Proposal • Automated B2B Lead Generation

Project Overview

LeadX Pro AI is a cutting-edge Lead Intelligence & Business Data Platform designed to automate the extraction, validation, and management of B2B leads. By leveraging advanced web scraping technologies and AI-powered Natural Language Processing (NLP), LeadX Pro AI transforms raw web data into verified, high-priority business contacts. The platform features a robust, distributed architecture allowing scalable concurrent scraping, real-time progress tracking, and seamless data exporting.

Problem Statement

Sales and marketing teams spend countless hours manually searching for leads, verifying contact information, and compiling data into spreadsheets or CRMs. Existing tools are often rigid, lack deep validation mechanisms, or struggle with complex, dynamic websites. This leads to low productivity, inaccurate data, and ultimately missed revenue opportunities. There is a critical need for an automated, intelligent system that can dynamically scrape targets, validate data on the fly, and prioritize high-quality leads.

Core Value Proposition

Automated Intelligence	Go beyond simple scraping with AI/NLP that intelligently identifies and extracts relevant business entities.
Deep Data Validation	Built-in pipelines automatically clean and verify emails (excluding disposable domains), phone numbers, and URLs, ensuring high confidence in lead quality.
Scalability & Performance	Built on a distributed task queue (BullMQ/Redis) capable of handling concurrent, resilient scraping jobs with automatic retries.
Seamless Exporting	Effortlessly filter and export curated lists into CSV, Excel, or JSON formats to plug directly into any CRM workflow.
Real-Time Visibility	Monitor scraping progress, success rates, and live data streaming directly via WebSockets.

System Architecture

Layer	Description
-------	-------------

Client Layer (Frontend)	React, Vite, Tailwind CSS SPA communicating via REST APIs and Socket.io for real-time updates.
API & Orchestration (Backend)	Node.js/Express server handling authentication, job scheduling, data validation, and database bridging.
Task Queue System	BullMQ backed by Redis for asynchronous processing, distributing tasks across multiple worker processes.
Scraping Engine	Headless browser environments (Puppeteer, Playwright) and HTML parsers (Cheerio) for distributed crawling and extraction.
Data Layer	MySQL 8.0+ as primary relational database ensuring ACID compliance for users, jobs, and leads.

Technology Stack

Frontend	React 19, Vite, Tailwind CSS, Zustand, React Query, Radix UI, Lucide React
Backend	Node.js (TypeScript), Express.js, Zod, Socket.io
Scraping & NLP	Puppeteer, Playwright, Cheerio, Natural (NLP), Compromise
Queueing & Cache	BullMQ, Redis
Database	MySQL 8.0+ via native drivers
Security	Helmet, CORS, Express Rate Limit, bcryptjs, JWT

Database Schema

Table	Purpose
users	User accounts, authentication, RBAC (admin/user)
scrape_jobs	Scraping tasks, configurations, target URLs, execution status
scraped_companies	Extracted lead data with verification statuses and confidence scores
exports	Data export requests, formats, and file generation status
scrape_history	Audit logs for job-related actions
saved_searches	User-defined filters for quick data retrieval
activity_logs	System-wide audit trails for security and monitoring
job_results	Page-level extraction results and error logs

Feature Specifications

1	Dashboard & Analytics Overview of scraped leads, job statuses, and system health metrics.
2	Job Management Create, pause, resume, and cancel scraping jobs with granular configuration (max pages, concurrency, extraction toggles).
3	Data Grid & Filtering Advanced data table with multifaceted filtering by industry, verification status, and lead priority.
4	Validation Pipeline Automated verification categorizing leads into verified, unverified, invalid, or pending based on data density and correctness.
5	Real-time Monitoring Live progress bars and streaming logs for active scraping jobs via WebSockets.
6	Export Engine Asynchronous generation of customized data exports in CSV, Excel, and JSON formats.

Scraping Engine — Technical Detail

The Scraping Engine is the core microservice responsible for data acquisition, built for resilience and high throughput.

Queue Architecture

Utilizes BullMQ with separate queues for **scrape**, **export**, and **retry** operations, enabling fine-grained control over job priorities and concurrency.

Job Configuration

Highly customizable parameters per job: **maxPages**, **concurrency**, **timeout**, preferred browser engine (Puppeteer / Playwright / Cheerio), and toggleable extraction modules (Email, Phone, Social Links).

Execution Flow

- A job is pushed to the scrapeQueue.
- A worker picks it up and launches the specified browser context.
- The engine navigates to the target, parses DOM elements, and applies NLP heuristics to locate entity data.
- Extracted raw data is passed through the Validation Pipeline, which sanitizes inputs (discarding disposable emails, standardizing phone formats, dropping spam business names).
- Validated data is bulk-inserted into scraped_companies with calculated confidence scores.

Resilience

Built-in retry mechanisms handle transient network errors or timeouts, pushing failed attempts to a retryQueue with exponential backoff, ensuring no data loss under adverse conditions.

